

Java Full Stack Developer

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SUMMARY:

- Detail-oriented Full Stack Developer with over 5+ years of experience in designing and developing enterprise-grade applications with expertise in:
- Advanced implementation of Collections framework, Generics, Multithreading, Exception Handling, and Garbage Collection tuning for optimized memory management.
- Strong expertise in implementing efficient data structures (Trees, Graphs, HashMaps, Heaps) and algorithms (Dynamic Programming, BFS/DFS, Divide and Conquer). Solved LeetCode problems across Easy, Medium, and Hard difficulties with optimal time and space complexity solutions.
- Extensive work with Java's concurrency utilities (Executors, ThreadPools, Futures, CompletableFuture), implementing thread-safe operations for high-throughput applications handling millions of daily transactions.
- Expertise in advanced multithreading concepts including thread synchronization, deadlock prevention, and non-blocking algorithm
- Proficient in implementing MVC, Singleton, Factory, Observer, Strategy, and Data Access Object patterns to create maintainable, extensible codebases.
- Expert in JVM tuning, profiling, and memory leak detection, having successfully optimized multiple production systems for improved response times and reduced resource consumption.
- Proven track record with Spring (Core, MVC, Boot, Security, Data), Hibernate, and RESTful/SOAP web services development.

Technical Skills:

Programming Languages	Java (Core, Collections, Concurrency, Data Structures & Algorithms), JavaScript
Java Frameworks	Spring (Core, Boot, MVC, Data, Security), Hibernate, JPA, JUnit, Mockito
Database	Oracle, PostgreSQL, MySQL, MongoDB, Cassandra
Cloud services	AWS (EC2, S3, Lambda, DynamoDB, RDS), Google Cloud Platform
Operating Systems	UNIX, Linux, Windows.
App Servers/ Middleware	Tomcat, WebLogic, JBoss, Apache Tomcat 7.x,.
Development Tools	IntelliJ, Eclipse, Git, Maven, Gradle
Query Languages	SQL, PL/SQL.
Web services	RESTful APIs, SOAP, GraphQL
Design patterns	MVC, Singleton, Factory, Data Access Object, Session Facade, Observer, Strategy
Tools:	JUnit, Log4j, ANT, Maven, JIRA.
Containerization	Docker, Kubernetes, Swarm

EXPERIENCE:

University Of Missouri Kansas City, Missouri

July 2024-June 2025

Full Stack Developer

- Implemented complex algorithms and data structures (AVL Trees, Graphs, Priority Queues) to optimize data processing pipelines, reducing search operations from $O(n)$ to $O(\log n)$

- Developed memory-efficient applications using advanced Java Collections framework, implementing custom data structures for specific business requirements
- Engineered concurrent processing modules using Java's ExecutorService and ThreadPoolExecutor to handle 10,000+ daily transactions with minimal latency
- Implemented thread-safe components using atomic operations, locks, and synchronized collections to prevent race conditions in high-concurrency environments
- Employed Java 8 Stream API with parallel processing for optimized data transformations, reducing processing time for large datasets
- Architected a highly concurrent multithreaded processing system using ReentrantLock, Semaphores, and CyclicBarriers to handle pharmaceutical compliance checks with precise thread coordination and non-blocking algorithms
- Architected microservices utilizing Spring Boot, applying Factory, Strategy, and Observer patterns to create maintainable and extensible codebase
- Implemented Repository and DAO patterns for database interactions, creating abstraction layers that improved testability and code organization
- Applied Builder pattern for complex object creation and Singleton pattern for managing shared resources across application components
- Designed event-driven architecture using the Observer pattern to decouple system components, improving scalability and maintainability
- Optimized JVM performance by fine-tuning garbage collection parameters, heap sizes, and implementing proper memory management practices
- Implemented connection pooling and query optimization techniques for MongoDB and PostgreSQL, reducing database response times for high-volume operations
- Created custom caching solutions using Java ConcurrentHashMap and Guava libraries to minimize redundant database calls
- Conducted regular code reviews and performance profiling to identify and eliminate memory leaks and optimize resource usage
- Led development of 15+ RESTful microservices exposing core business functionality with robust error handling and validation
- Implemented OAuth2 with JWT for secure authentication and authorization across service boundaries
- Containerized applications with Docker and orchestrated with Kubernetes for improved deployment reliability and scalability
- Established CI/CD pipelines using Jenkins, automating testing and deployment workflow.

Acentra Health, Rockville MD

Dec 2022- June 2024

Java Full Stack Developer

- Applied algorithmic techniques from LeetCode problem solving to optimize data processing pipelines, implementing efficient sorting and searching algorithms that reduced processing time by 25%
- Designed and implemented thread-safe components for a Tax Filing and Reporting System processing concurrent user requests
- Utilized Java's concurrent collections (ConcurrentHashMap, CopyOnWriteArrayList) to manage shared data structures across multiple threads
- Applied CompletableFuture API for asynchronous processing of financial calculations and report generation

- Optimized memory usage by implementing proper object lifecycle management and minimizing object creation in critical paths
- Employed Factory and Builder patterns for creating complex financial document objects with multiple configuration options
- Implemented Observer pattern to notify relevant system components of tax filing status changes
- Applied Decorator pattern to extend functionality of core reporting components without modifying existing code
- Used Repository pattern with Spring Data JPA to abstract and simplify database operations
- Enhanced PostgreSQL database performance through schema optimization and query tuning for a system handling millions of daily transactions
- Implemented thread pool configurations that balanced throughput and resource utilization based on production workload patterns
- Applied batch processing strategies using Spring Batch for efficient bulk operations, reducing processing time by 40%
- Conducted JVM profiling to identify and resolve memory leaks in long-running batch processes
- Developed secure user management with role-based access control using Spring Security
- Created responsive user interfaces using React and jQuery for improved user experience
- Deployed cloud infrastructure on Microsoft Azure, optimizing performance and cost efficiency
- Established CI/CD pipelines with Jenkins for automated testing and deployment.

Accenture Pvt Ltd | Software Developer

July 2021- July 2022

Role: Java / J2EE Developer

Responsibilities:

- Leveraged Java's concurrent utilities (CountDownLatch, CompletableFuture) to orchestrate complex, interdependent background tasks
- Implemented custom thread-safe data structures and producer-consumer patterns for efficient data processing
- Applied ThreadLocal variables to maintain thread safety in stateful processing contexts
- Utilized atomic classes and concurrent collections to ensure data consistency in multithreaded environments
- Designed microservices architecture following Domain-Driven Design principles with clearly defined bounded contexts
- Implemented Adapter pattern to integrate with legacy systems and third-party APIs
- Applied Chain of Responsibility pattern for flexible request processing pipelines
- Used Command pattern to encapsulate business operations, enabling features like transaction management and operation logging
- Optimized Oracle SQL database interactions through proper indexing strategies and query tuning
- Implemented connection pooling and statement caching to reduce database connection overhead
- Applied lazy loading and pagination techniques to improve response times for large data sets
- Conducted regular performance testing to identify and resolve bottlenecks in production systems
- Secured authentication using Spring Security and JWT for API access control

- Utilized atomic classes and concurrent collections to ensure data consistency in multithreaded environments.
- Led front-end development using React.js, creating responsive web applications
- Implemented Kubernetes for container orchestration, improving deployment efficiency
- Reduced manual QA effort through comprehensive JUnit and Mockito test automation.

Picon Infotech , India

Jan 2020 - June2021

Java Developer

- Developed reusable utility classes implementing Java Collections framework for efficient data manipulation
- Created multithreaded components for processing user requests concurrently while maintaining data consistency
- Implemented thread pooling strategies to optimize resource usage in high-traffic scenarios
- Applied Java 8 Stream API and Lambda expressions to streamline data processing workflows
- Implemented MVC architecture using Spring MVC for web application development
- Applied DAO pattern with Hibernate for database operations, separating business logic from data access
- Utilized Singleton and Factory patterns for managing shared resources and object creation
- Designed RESTful APIs following best practices for endpoint naming and resource representation
- Enhanced application performance through proper connection pooling and resource management
- Implemented efficient Hibernate query strategies to minimize database load
- Applied caching mechanisms to reduce redundant computations and database queries
- Optimized JVM memory usage through careful object lifecycle management
- Created single-page applications using Angular 8 and JavaScript
- Developed Docker containers orchestrated by Kubernetes on Google Cloud Platform
- Implemented robust data persistence using Hibernate and Spring Framework
- Created PL/SQL stored procedures for optimized database operations

Education

Gitam university,India

July 2017-May 2021

Bachelor of Engineering in Computer Science

Univeristy of Missouri-kansas city

Aug 2022-May2024

master's in computer science



